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DATA NOTE

The Danish Lymphoid Cancer Research (DALY-CARE): Genetic Cohort Profile

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Purpose: The Danish Lymphoid Cancer Research (DALY-CARE) Genetic Cohort was established to support research into how genetic factors influence clinical outcomes in lymphoid cancers (LCs), including disease progression, treatment response, toxicity, and survival. Individual-level genetic data were combined with detailed clinical information from national health registers, hospital-based electronic health records (EHR), laboratory data, and pathology reports. The cohort enables large-scale studies of genetic susceptibility, disease course, and therapy-related outcomes in LCs and provides a platform for genetic epidemiology and future multi-omics research within a unified data infrastructure.

Participants: The genetic cohort includes 8675 genotyped individuals drawn from the broader DALY-CARE population (n=74,251, as of April 2025), including individuals diagnosed with LCs such as diffuse large B-cell lymphoma (DLBCL, n=1349), chronic lymphocytic leukemia (CLL, n=1245), multiple myeloma (MM, n=1209), follicular lymphoma (FL, n=704), Hodgkin lymphoma (HL, n=407), Waldenström macroglobulinemia and lymphoplasmacytic lymphoma (WM/LPL, n=368), marginal zone lymphoma (MZL, n=287), and precursor states such as monoclonal gammopathy of undetermined significance (MGUS, n=1299).

Descriptive Findings: Hematologic malignancy was the most frequent cause of death (30%), followed by infections (27%) and other cancers (15%). Polypharmacy, as a more sensitive proxy for comorbidity than hospital diagnosis codes, was substantial (median 7–8 drugs pre-diagnosis). Frequently observed comorbidities were hypertension (47%), cardiovascular disease (16%), cerebrovascular disease (11%) and type 2 diabetes (10%). Kinship analysis identified limited relatedness (41 parent–offspring, 51 siblings), while ancestry inference confirmed predominantly Northwestern European descent (97%).

Future Opportunities: The DALY-CARE Genetic Cohort provides a foundation for studying genetic and clinical determinants of LC outcomes. Integration of genotype data with EHR and national health registers enables exploration of germline risk and protective variants. Future expansions will integrate additional omics data types, such as whole-genome sequencing, transcriptomics, proteomics, and immunophenotyping, positioning the cohort as a national platform for multi-omics research in LC.

Keywords: haematology, lymphoma, myeloma, leukemia, genetics, epidemiology

Introduction

Lymphoid-lineage cancers (LCs) cover the heterogeneous group of chronic lymphocytic leukemia (CLL), lymphomas, and multiple myeloma (MM) and their precursors. After marked incidence increases during the late 20th century,[1] LCs

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now rank among the ten most common cancers globally and account for approximately 10% of all incident cancer cases each year.[2–4] Although biologically distinct, they share epidemiological patterns, lymphoid lineage origin and treatment approaches.[5–7] Studying them together can therefore reveal both shared and subtype-specific disease mechanisms and outcomes.

LC patients are often older and live with multiple comorbidities and polypharmacy,[8] all of which influences treatment choices, treatment toxicity, and survival. Randomized clinical trials often underrepresent this complexity due to strict inclusion- and exclusion criteria,[9] leaving key questions unanswered about treatment effectiveness, survivorship, and the biological basis of patient-to-patient variation. Combining comprehensive real-world clinical data with germline genetic data allows these questions to be addressed directly. Genetic data can identify inherited variants that modify treatment response, toxicity, and long-term outcomes, while real-world data capture the clinical context in which those effects occur. Together, these data enable evaluation of how inherited variation shapes disease trajectories and may inform more precise, individualized treatment strategies.

The Danish Lymphoid Cancer Research (DALY-CARE) Genetic Cohort was established to meet the need for such linked genetic and real-world data. It combines individual level genotype data for 8675 individuals with detailed clinical data from national health registries and hospital-based electronic health records (EHR), including treatment histories, laboratory values, pathology reports, as well as longitudinal follow-up. This enables studies of inherited genetic variation in LCs, and research into how germline genetics and clinical factors influence outcomes across the entire disease trajectory of LCs.

The DALY-CARE Genetic Cohort is nested within the broader DALY-CARE data resource, which includes all patients diagnosed with LCs in Denmark since 2002 (n=74,251, as of April 2025).[10] The DALY-CARE resource has proven valuable in translating data-driven research into clinical practice. This is exemplified by the implementation of the CLL-TIM risk model into the EHR system of Eastern Denmark to support decision-making and identify newly diagnosed patients with high-risk CLL for clinical trials.[11–13] This demonstrates that data-driven models derived from DALY-CARE can be translated directly into clinical practice. Integrating germline genetic data into this infrastructure is a logical next step to improve prediction of disease trajectories and guiding personalized treatment.

Cohort Description

Population Characteristics

We included individuals fulfilling the following four criteria: 1) adult residents of Denmark, 2) enrolled in DALY-CARE, 3) registered with at least one LC diagnosis since January 1, 2002, (Table 1 and Supplemental Table S1), and 4) with available biobank samples in one of four Danish biobanks (detailed in *Samples* section). This currently comprises 8675 individuals (56% males). The median age at first diagnosis for the included individuals was 70 years (interquartile range [IQR] 62.0;77.0) for males and 71 years (IQR 62.1;78.0) for females. Their diagnoses comprise a heterogeneous group of hematological malignancies and malignancy precursor states (Table 1), including diffuse large B-cell lymphoma (DLBCL, n=1349), monoclonal gammopathy of undetermined significance (MGUS, n=1209), chronic lymphocytic leukemia (CLL, n=1245), multiple myeloma (MM, n=1299), follicular lymphoma (FL, n=704), Hodgkin lymphoma (HL, n=407), Waldenström's macroglobulinemia and lymphoplasmacytic lymphoma (WM/LPL, n=368), T-cell lymphomas (n=342), marginal zone lymphoma (MZL, n=287), small lymphocytic lymphoma (SLL, n=218), mantle cell lymphoma (MCL, n=197), smoldering multiple myeloma (SMM, n= 59), hairy cell leukemia (HCL, n=58), monoclonal B-cell lymphocytosis (MBL, n=41), Burkitt lymphoma (BL, n=32), lymphoblastic lymphoma (LBL, n=27), plasma cell leukemia (PCL, n=9), and other unspecified leukemias and lymphomas (n=824).

Register-Based Data

The DALY-CARE data resource is a comprehensive collection of longitudinal clinical data primarily sourced from existing Danish national registers, hospital-based EHRs, and laboratory systems. The data sources integrated into DALY-

Table 1 Basic Characteristics of the DALY-CARE Genetics Cohort

Characteristics	Male (n = 4869)	Female (n = 3806)	Total n. Values (n = 8675)
Sociodemographics			
Age at diagnosis, median (IQR)	70.1 (61.6; 77.3)	70.9 (61.7; 78.4)	8675
Smoking status (at diagnosis)[a], n (%)	3008	2438	5446
Never smoker	1008 (33.5)	1091 (44.7)	2099 (38.5)
Former or passive smoker	1456 (48.4)	956 (39.2)	2412 (44.3)
Current smoker	544 (18.1)	391 (16.0)	935 (17.2)
Follow-up time (from first diagnosis)			
Follow-up time in years, median (IQR)	5.9 (2.5; 9.7)	6.8 (3.1; 10.3)	8675
Anthropometry (at diagnosis)			
Body mass index (kg/m2), median (IQR)[a]	25.8 (23.4; 28.7)	24.6 (21.6; 28.3)	6269
Lymphoid cancer diagnosis (DHQI or first diagnosis)[b]			
Diffuse Large B-cell Lymphoma (%)	755 (15.5)	594 (15.6)	1349 (15.6)
Monoclonal Gammopathy of undetermined significance (%)	707 (14.5)	592 (15.6)	1299 (15.0)
Chronic Lymphocytic Leukemia (%)	758 (15.6)	487 (12.8)	1245 (14.4)
Multiple Myeloma (%)	652 (13.4)	557 (14.6)	1209 (13.9)
Follicular Lymphoma (%)	326 (6.7)	378 (9.9)	704 (8.1)
Hodgkin Lymphoma (%)	224 (4.6)	183 (4.8)	407 (4.7)
Waldenström Macroglobulinemia, Lymphoplasmacytic Lymphoma (%)	239 (4.9)	129 (3.4)	368 (4.2)
T-Cell Lymphoma (%)	216 (4.4)	126 (3.3)	342 (3.9)
Marginal Zone Lymphoma (%)	127 (2.6)	160 (4.2)	287 (3.3)
Small Lymphocytic Lymphoma (%)	133 (2.7)	85 (2.2)	218 (2.5)
Mantle Cell Lymphoma (%)	127 (2.6)	70 (1.8)	197 (2.3)
Smoldering Multiple Myeloma (%)	40 (0.8)	19 (0.5)	59 (0.7)
Hairy cell leukemia (%)	47 (1.0)	11 (0.3)	58 (0.7)
Monoclonal B-cell Lymphocytosis (%)	21 (0.4)	20 (0.2)	41 (0.5)
Burkitt lymphoma (%)	20 (0.4)	12 (0.3)	32 (0.4)
Lymphoblastic lymphoma	21 (0.4)	6 (0.2)	27 (0.3)
Plasma cell leukemia	7 (0.1)	2 (0.1)	9 (0.1)
Unspecified Lymphoma (%)	312 (6.4)	285 (7.5)	597 (6.9)
Unspecified Leukemia (%)	105 (2.2)	77 (2.0)	182 (2.1)
Other (%)	32 (0.7)	13 (0.3)	45 (0.5)

Notes: [a]Number of patients with missing information: smoking: NA=3229; Body mass index: NA=2406. [b]Included disease entities with their respective ICD-10 codes are listed in Supplemental Table S1. Diagnoses were obtained in the following order: first from the DHQI registry, if unavailable, the first recorded diagnosis was used.

Abbreviations: IQR, interquartile range; NA, not available; ICD-10, International Classification of Diseases, 10th Revision; DHQI, the Danish Healthcare Quality Institute.

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CARE have been previously described.[10] In brief, they include: 1) the Danish Healthcare Quality Institute (DHQI, formerly known as the Danish Clinical Quality Program – National Clinical Registries [RKKP]);[14–16] 2) Danish Health Data Authority (SDS);[17] and 3) EHR including data from 16 modules of the EPIC®-based EHR system (Sundhedsplatformen [SP]) in eastern Denmark.

Laboratory Data

Laboratory data is available from the Clinical Laboratory Information System Database. *TP53* aberration analysis results are available for 1763 patients. Immunoglobulin heavy chain variable (IGHV) mutational status, including genetic sequence and stereotypic subset analyses in CLL/SLL, is available for more than 1000 cases and immunoglobulin heavy change (IGH) split status is available for more than 700 patients with MM within the DALY-CARE genetic cohort. Lastly, fluorescence in situ hybridization (FISH) results on patients with CLL and MM are available from DHQI registers, and 190 unique molecular International Classification of Diseases for Oncology, Third Edition (ICD-O-3) and Systematized Nomenclature of Medicine (SNOMED) codes related tests for other specific genetic mutations (prefix "FE") are available for 934 patients within the pathology register (Supplemental Table S2).

Samples

Biobank samples from individuals in DALY-CARE were obtained from four major Danish biobanks:[18] namely, the CLL Biobank,[10] the Copenhagen Hospital Biobank,[19] the Danish Cancer Biobank,[20] and the PERSIMUNE Biobank.[21]

Follow-up

Patients in DALY-CARE are continuously followed through the linked Danish national registers and EHRs. All DHQI, SDS, and EHR data are updated annually at minimum, with the EHR data also available for updates upon request. This allows for the compilation of data on survival, long-term temporal registration of hospital admissions, procedures, and treatments. The continuous and timely updating of these datasets enables research into new treatments and clinical procedures, allowing for rapid assessment of their implementation, efficacy, and outcomes in parallel with their introduction into clinical practice. The median time from first diagnosis to end of follow-up (April 2025) or death is 5.9 years (IQR: 2.5, 9.7) for males and 6.8 years (IQR: 3.1, 10.3) for females (Table 1). The cohort includes patients diagnosed as early as 2002, the average follow-up duration reflects the high mortality of this patient population rather than limited observation time.

Data Storage

All DALY-CARE data are housed and processed within the secure computational infrastructure of the Danish National Genome Center (NGC), which operates under ISO 27001 certification.[22] To safeguard individual patient privacy, records are pseudonymized by assigning each record a unique DALY-CARE identifier, which is cross-referenced with the Danish Civil Registration System to allow consistent linkage across health and administrative registers.

The data resource is organized into two complementary layers: an import layer, which retains raw (but pseudonymized) incoming files, and a core layer, which contains harmonized and aggregated datasets. Standardized lookup tables ensure interoperability across coding frameworks such as SNOMED, the Danish Health Classification System (SKS), the Anatomical Therapeutic Chemical Classification System (ATC), the International Classification of Diseases, 10th Revision (ICD-10), and Nomenclature for Properties and Units (NPU). These technical specifications, together with the database architecture, are documented in detail in the DALY-CARE data resource paper.[10] Data integration and transformation are performed through reproducible open-source workflows implemented in R, PostgreSQL and Python.

Patient and Public Involvement

A planned study examining adverse events following treatment for lymphoma and MM using the DALY-CARE Genetic Cohort was presented to the Patient and Relatives Council at the Danish Cancer Society. The direct engagement of patients (or their relatives) ensured that the research questions and focus areas were aligned with the lived experiences

and priorities of the target patient population. Patient organizations and patient representatives have been involved to ensure representation of their perspective in research projects such as the planned expansion of DALY-CARE to include all hematological malignancies, as well as a control population.

Genetics

As part of a collaboration with deCODE Genetics (Reykjavík, Iceland), available biobank peripheral whole blood samples from patients included in DALY-CARE were shipped to deCODE for DNA extraction, genotyping, and subsequent imputation.

Variant Imputation

Genotype imputation was carried out by deCODE Genetics using a combined reference panel: (1) the UK 1000 Genomes Phase 3 and HapMap datasets to impute common variants with a minor allele frequency (MAF) above 1%, and (2) a proprietary reference set of more than 6000 Danish whole-genome sequences to improve the imputation accuracy of rare variants down to a MAF of approximately 0.01%.[23]

Copy Number Variations

Copy number variations (CNVs) were identified by deCODE Genetics using single nucleotide polymorphism (SNP) array intensity data, specifically B-allele frequency (BAF), which reflects the proportion of signal attributable to the B allele at each SNP and log R ratio (LRR), which measures the total intensity of the signal compared to what is expected for a normal diploid genome based on reference intensity values. CNV calling was performed using the PennCNV.[24] Variants supported by fewer than 20 SNPs were excluded to reduce false positives. Remaining CNVs underwent manual quality control (QC), including visual inspection of BAF and LRR plots, to further exclude artifacts.

Quality Control

QC was performed using PLINK (v1.9 and v2.0) following established best practices for genotype data.[25–27] SNPs and individuals with >20% missingness were removed in an initial filtering step, followed by exclusion of those with >2% missingness. Samples with discordant genetic and reported sex, assessed via X chromosome inbreeding coefficients, were excluded (n=34). Variants with an in-cohort minor allele frequency (MAF) <1% and those deviating from Hardy-Weinberg equilibrium ($p < 10^{-5}$) were excluded. Heterozygosity outliers (>3 standard deviations [SD] from the mean) were then removed. This resulted in a final dataset of 8675 individuals and 11,565,053 variants. A flowchart summarizing the QC steps is provided in Supplemental Figure S1. The post-QC MAF distribution is shown in Figure 1.

Descriptive Findings

Generalizability of the DALY-CARE Genetic Cohort

Patients with the nine most prevalent diagnoses in the DALY-CARE Genetic Cohort including CLL, DLBCL, FL, MM, MZL, T-cell lymphomas, and WM/LPL (n=7210) were compared with patients with the same nine diagnoses from the broader DALY-CARE cohort (n=54,721) to assess the generalizability of the genotyped cohort. Kaplan–Meier analyses showed broadly similar overall survival (OS) between genotyped and non-genotyped patients for most diagnoses (Figure 2). Shorter OS was observed for genotyped patients with HL and MGUS compared to corresponding non-genotyped patients. This likely reflects selection bias, as genotyping was contingent on prior hospital contact and blood typing.[19] Non-genotyped patients with HL and MGUS were significantly younger (Supplemental Figure S2), reflecting that younger patients with HL or MGUS were less likely to have a blood type taken, thus more likely to be included in the non-genotyped part of the cohort, with age potentially contributing to their relatively favorable outcomes. The distribution of HL-specific International Prognostic Index (IPI) was broadly similar between genotyped and non-genotyped patients (Supplemental Figure S3). When assessing other lymphoma subtype-specific IPI, genotyped patients with MM and FL appeared to have a higher proportion of high-risk IPI scores compared with non-genotyped patients.

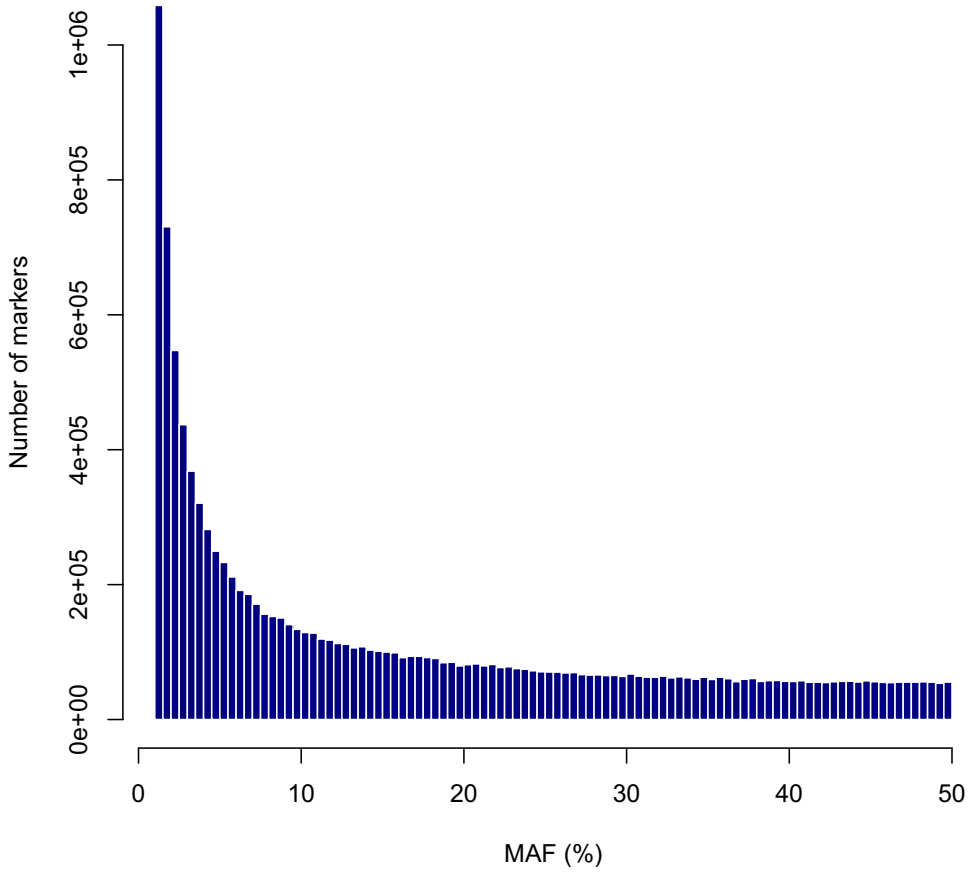


Figure 1 Distribution of minor allele frequencies (MAF) of genotyped single nucleotide polymorphisms (SNPs) after quality control.

Overall, the findings suggest that, for most diagnoses, genotyping status does not influence survival and reflects random sampling from the broader DALY-CARE population, with differences in HL and MGUS primarily driven by selection bias related to patterns of clinical care and hospital presentation rather than genotyping itself.

Comorbidities, Polypharmacy and Causes of Death

IPI, Eastern Cooperative Oncology Group [ECOG] performance status, and Charlson Comorbidity Index for selected LCs are presented in Table 2. Comorbidity assessed through medication use was substantial: polypharmacy was common, with a median of 7–8 distinct prescription drugs (using the seventh ATC level) within the last year prior to diagnosis (Table 3 and supplemental Table S3). This aligns with recent work performed within the full DALY-CARE data resource, where pre-diagnostic polypharmacy was shown to be highly prevalent and an independent predictor of survival, hospitalization, and severe infections in lymphoid cancers.[8] Frequently observed comorbidities were hypertension, affecting 4084 patients (47%), cardiovascular diseases (16%), cerebrovascular disease (11%), and type 2 diabetes (10%). Hypertension and type 2 diabetes were defined using combined ICD-10 diagnoses and prescription-based criteria as applied by Rotbain et al,[28] while all other comorbidities were defined using ICD-10 codes alone (Supplemental Table S4) These prevalences differ from crude estimates in the general Danish population, where cardiovascular disease affects ~11%, chronic kidney disease ~9.5%, and type 2 diabetes ~4.5%.[29–31] However, the apparent excess burden is largely attributable to the older age distribution of the DALY-CARE genetic cohort. For type 2 diabetes, previous work has demonstrated that the prevalence in CLL patients is comparable to that of age-matched individuals without CLL.[28]

Taken together, the substantial burden of comorbidities and extensive polypharmacy patterns demonstrate that the cohort is well-powered to study LC in the context of significant comorbid conditions, reflecting the real-world complexity of managing these malignancies in an aging population.

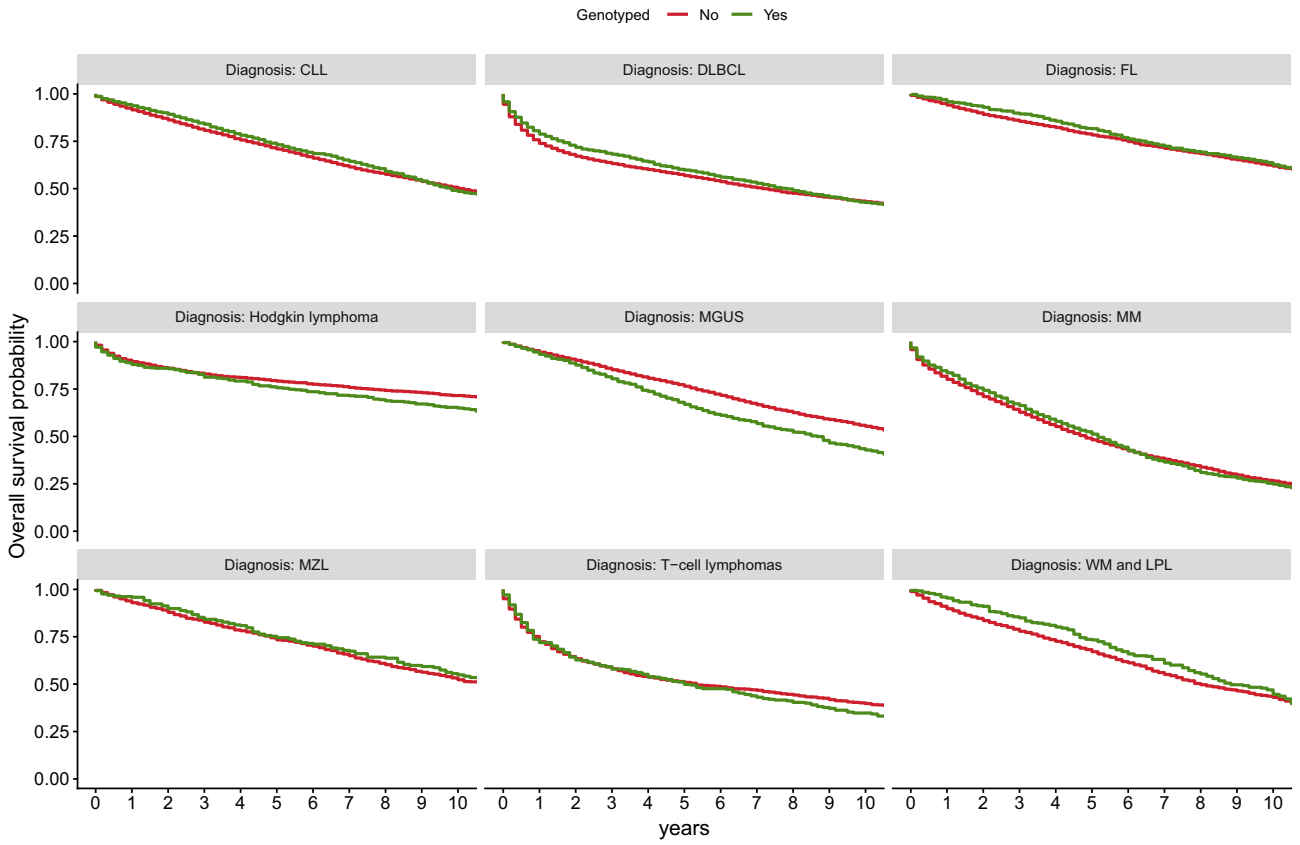


Figure 2 Kaplan-Meier plots depicting overall survival by genotyping status, grouped by diagnosis (Log rank test) for Chronic lymphocytic leukemia (CLL), Diffuse large B-cell lymphoma (DLBCL), Follicular lymphoma (FL), Hodgkin lymphoma (HL), Monoclonal gammopathy of undetermined significance (MGUS), Multiple myeloma (MM), Marginal zone lymphoma (MZL), Waldenström's macroglobulinemia (WM), Lymphoplasmacytic lymphoma (LPL).

The most common underlying cause of death was hematological malignancy (N=1237, 30%), followed by infections (n=1112, 27%) and other cancers (n=632, 15%) (Table 3, Supplemental Table S5).

Kinship

We assessed relatedness using pairwise identity-by-descent (IBD) estimation based on genome-wide SNP data. Pairwise relatedness was estimated using PLINK v1.9 to calculate IBD sharing statistics.[26] Genotype data were pruned for linkage disequilibrium, and autosomal variants with a MAF $\geq$ 0.01 were retained. Individuals with genome-wide relatedness estimates (PI_HAT) greater than 0.2 were considered potentially related. Their relationships were further characterized through Z0 (proportion of the genome where no alleles are shared IBD) and Z1 (one allele shared), and Z2 (two alleles shared) statistics. Relationship categories, including monozygotic twins, parent-offspring, full siblings, second-degree relatives, and third-degree relatives, were inferred based on expected IBD patterns and PI_HAT thresholds as part of the quality control for the cohort (Supplemental Table S6).[32–34] We identified no monozygotic twin pairs, 41 parent-offspring pairs, 51 full sibling pairs, 90 second-degree relatives, and 281 third degree relatives, information that may also provide insights into impact of familiar versus non-familiar LC (Figure 3).[35]

Ancestry

Ancestry was inferred by projecting individuals onto a reference principal component (PC) space constructed from the UK Biobank and 1000 Genomes populations. We used the pre-computed PC loadings provided by Privé et al, applying the recommended shrinkage correction to avoid overfitting.[36,37] Reference population centers and ancestry groups were taken from the UK Biobank-derived global ancestry reference defined by Privé et al[38] After SNP matching and removing variants with >5% missingness, individual genotype data were projected onto this reference PC space. PC loadings were examined to confirm

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Table 2 Performance Status, Comorbidity, and Prognostic Indices at Diagnosis

Characteristics	Disease								Total n. Values	NA[a]
	DLBCL	CLL	MM	FL	HL	LPL/WM	MZL	MCL		
ECOG Performance Status at diagnosis, median [IQR]	1 [0, 1]	0 [0, 0]	1 [0, 1]	0 [0, 1]	0 [0, 1]	0 [0, 1]	0 [0, 1]	0 [0, 1]	4951	3724
Charlson Comorbidity Index at diagnosis, median [IQR]										
Median [IQR]	3 [2, 4]	2 [2, 3]	2 [2, 4]	2 [2, 3]	2 [2, 3]	2 [2, 4]	2 [2, 3]	2 [2, 3]	8454	221
0, n(%)	74 (5.4)	84 (7.0)	115 (9.8)	40 (6.0)	31 (8.6)	18 (6.4)	15 (3.8)	10 (5.2)		
1, n(%)	37 (2.8)	37 (3.1)	54 (4.6)	17 (2.0)	14 (3.9)	10 (3.6)	7 (1.8)	4 (2.1)		
2, n(%)	546 (41.5)	532 (44.1)	442 (37.5)	306 (45.9)	141 (39.2)	119 (42.3)	212 (54.4)	98 (50.8)		
≥3, n(%)	660 (50.1)	552 (45.8)	567 (48.1)	304 (45.6)	174 (48.3)	134 (47.7)	156 (40.0)	81 (42.0)		
International Prognostic Index at diagnosis[b], n	1349	1245	1209	704	407	368	287	197	5766	2909
Low, n (%)	81 (6.0)	347 (27.9)	82 (6.8)	73 (10.4)	158 (38.8)	45 (12.2)	56 (19.5)	33 (16.8)		
Intermediate, n (%)	588 (43.6)	196 (15.7)	505 (41.8)	342 (48.6)	–	155 (42.1)	114 (39.7)	72 (36.5)		
High, n (%)	569 (42.2)	109 (8.8)	125 (10.3)	150 (21.3)	130 (32.0)	68 (18.5)	99 (34.5)	79 (40.1)		
NA[a], n (%)	111 (8.2)	593 (47.6)	497 (41.1)	139 (19.7)	119 (29.2)	100 (27.2)	18 (6.3)	13 (6.6)		

Notes: [a]NA: number of patients with missing information. [b]International Prognostic Index (IPI) represents R-IPI in DLBCL, CLL-IPI in CLL (very high and high risk reported as high risk), R-ISS in MM, FLIPI2 in FL, IPSSWM in LPL/WM, MALT-IPI in MZL, IPS in HL, and MIPI in MCL.

Abbreviations: ECOG, Eastern Cooperative Oncology Group; IQR, interquartile range; DLBCL, Diffuse large B-cell lymphoma; CLL, Chronic lymphocytic leukemia; MM, Multiple myeloma; FL, Follicular lymphoma; LPL/WM, Lymphoplasmacytic lymphoma / Waldenström's macroglobulinemia; MZL, Marginal zone lymphomas; HL, Hodgkin lymphoma; MCL, Mantle cell lymphoma; NA, not available.

Table 3 Comorbidities, Treatment Regimens, and Causes of Death

Characteristics	Male (n = 4869)	Female (n = 3806)	Total n. Values (n = 8675)
Comorbidities (prior to first diagnosis)[a]			
Number of distinct ATC codes in the year prior to first diagnosis, median (IQR)	7 (4–11)	8 (4–12)	8093
Hypertension, n(%)	2344 (48.1)	1740 (45.7)	4084 (47.1)
Type 2 Diabetes, n(%)	576 (11.8)	280 (7.4)	856 (9.9)
Cardiovascular diseases, n(%)	943 (19.4)	444 (11.7)	1387 (16.0)
Cerebrovascular disease, n(%)	608 (12.5)	362 (9.5)	970 (11.2)
Liver disease, n(%)	119 (2.4)	104 (2.7)	223 (2.6)
Chronic kidney disease, n(%)	312 (6.4)	144 (3.8)	456 (5.3)
First-line treatment[b]			
Chemotherapy only, n(%)	887 (18.2)	666 (17.5)	1553 (17.9)
Chemoimmunotherapy (e.g, R-CHOP), n(%)	1340 (27.5)	1004 (26.4)	2344 (27.0)
Targeted therapy (e.g. BTKi inhibitors), n(%)	110 (2.3)	66 (1.7)	176 (2.0)
Proteasome inhibitor-based therapy, n(%)	326 (6.7)	249 (6.5)	575 (6.6)
Immunomodulatory drugs, n(%)	74 (1.5)	80 (2.1)	154 (1.8)
Stem cell transplant, n(%)	80 (1.6)	57 (1.5)	137 (1.6)
Other, n(%)	219 (4.5)	188 (4.9)	407 (4.7)
Watch and wait[c], n(%)	1833 (37.6)	1496 (39.3)	3329 (38.4)
Cause of death[d]	(n=2437)	(n=1668)	(n=4105)
Hematological/lymphatic malignancies, n(%)	703 (28.8)	534 (32.0)	1237 (30.1)
Infections, n(%)	677 (27.8)	435 (26.1)	1112 (27.1)
Other malignancies, n(%)	369 (15.1)	263 (15.8)	632 (15.4)
Cardiovascular causes, n(%)	192 (7.9)	88 (5.3)	280 (6.8)
Cerebrovascular causes, n(%)	29 (1.2)	25 (1.5)	54 (1.3)
Other causes, n(%)	467 (19.2)	323 (19.4)	790 (19.2)

Notes: [a]Included disease entities with their respective ICD-10 codes are listed in Supplemental Table S4. [b]Treatment regimens included in each category are listed in Supplemental Table S3. [c]Defined as patients having not received first-line treatment at end of follow-up. [d]Included disease entities with their respective ICD-10 codes are listed in Supplemental Table S5. Number of patients with missing information: NA=4570.

Abbreviations: ATC, Anatomical Therapeutic Chemical; BTKi, Bruton tyrosine kinase inhibitor; ICD-10, International Classification of Diseases, 10th Revision; IQR, Interquartile range; NA, Not available; R-CHOP, Rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone.

balanced contribution across the genome (Supplemental Figure S4). Squared Euclidean distances to reference population centers in this PC space were then computed, and individuals were assigned to the nearest ancestry group if within a predefined distance threshold. The majority of participants clustered with individuals of European ancestry (North West n=8432, 97.2%, South East n=80, 0.9%, North East n=76, 0.9%, South West n=16, 0.2%), followed by those of Ashkenazi (n=47, 0.5%), Finnish (n=17, 0.2%) and South American (n=5, 0.1%) ancestries (Figure 4 and Supplemental Table S7). Compared with nationwide Danish registry studies of hematological malignancies, which report a substantially higher proportion of patients of non-Western origin, this suggests an under-representation of non-Western ancestry in the genotyped subset of the cohort.

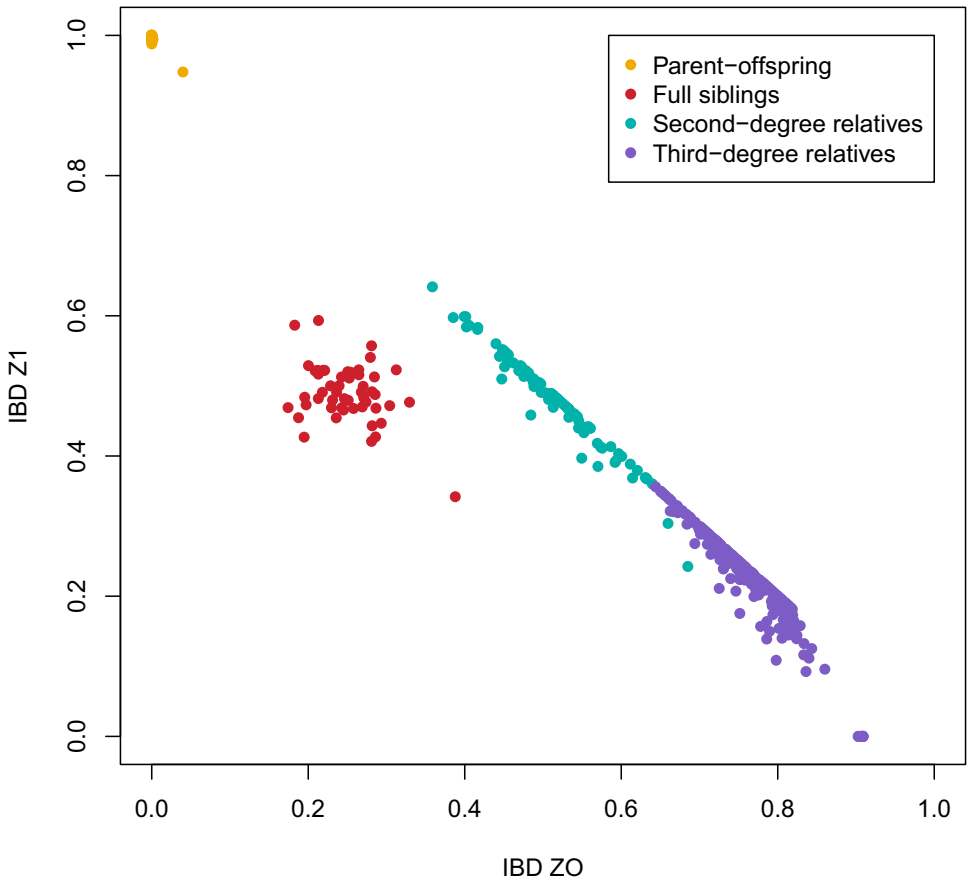


Figure 3 Pairwise identity-by-descent (IBD) sharing in the cohort. The plot shows the proportion of single nucleotide polymorphisms (SNPs) where each pair of individuals shares 0 (Z0) or 1 (Z1) allele. Points are colored by inferred relationship type: parent-offspring, full siblings, second-degree, and third-degree relatives.

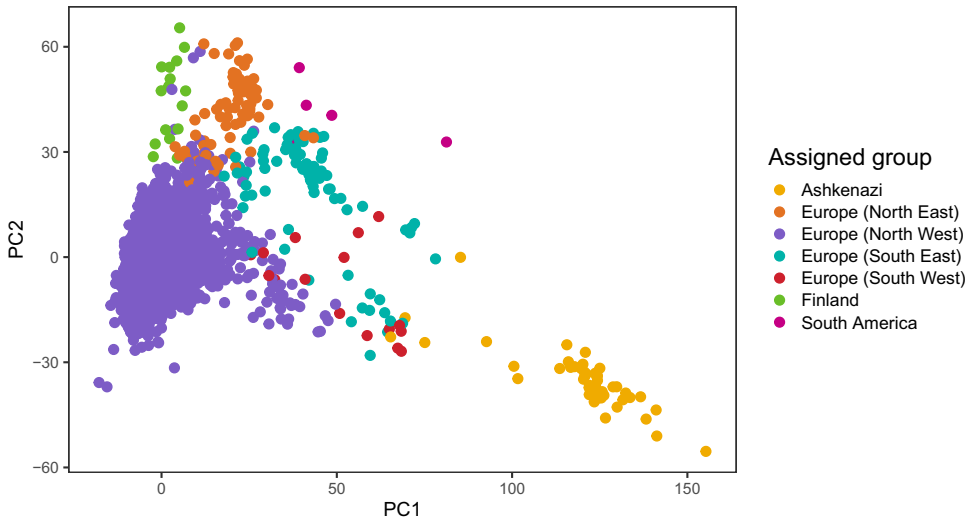


Figure 4 Principal component analysis of genetic ancestry in the DALY-CARE Genetic Cohort. The first two principal components (PC1 and PC2) are shown, with each point representing an individual. Colors indicate assigned ancestry groups: Ashkenazi, Europe (North East, North West, South East, South West), Finland, and South America.

Strengths and Limitations

A key strength of the DALY-CARE Genetic Cohort is that it is embedded within the nationwide, population-based DALY-CARE data resource, which captures nearly all cases of LC in Denmark due to mandatory cancer registration. This provides near-complete coverage of the Danish LC population and supports robust, real-world analyses across disease subtypes and treatment settings. The genetic cohort benefits from high-quality genotyping data, and linkage to rich longitudinal clinical data from multiple national health registers, hospital-based EHRs, and laboratory data.

Comprehensive hospital-based EHR data are available only from the eastern part of Denmark (covering approximately half of the 6 million Danish population). Additionally, the use of ICD-10 codes assigned at hospital admissions may vary across institutions, and shifts in data systems (eg, transitions between versions of Danish National Patient Register in February 2019, and the implementation of the EPIC®-based EHR in eastern Denmark from May 2016 to November 2017) may introduce time-dependent changes in data and formats, which must be carefully accounted for to avoid potential biases.

A limitation is that germline genotypes were derived from peripheral blood, which in leukemic malignancies such as CLL and PCL may contain circulating tumor cells. While copy number calling and quality control mitigate the effects of large deletions and duplications, copy-neutral loss of heterozygosity and low-level subclonal allelic imbalance may not be fully captured by copy number–based filters.[39] Consequently, contamination of blood-derived germline DNA by malignant cells cannot be completely ruled out, and results should be interpreted accordingly.

The cohort lacks population-based controls, limiting some types of genetic association studies. Finally, this data resource is unlikely to be of use for the study of dependence of associations on ethnicity, due to the small numbers of patients of non-Northwestern ethnicity.

Future Perspectives

The DALY-CARE Genetic Cohort is uniquely positioned to examine the impact of germline genetics on the clinical progression of LCs, treatment response, and adverse events, including risk of toxicity and late effects upon different treatment regimens, in a real-world population. The data resource includes linkage of germline genetic profiles with detailed, longitudinal clinical data. The cohort continues to grow, with ongoing genotyping of over 3000 additional individuals, as well as the planned integration of whole-genome sequencing data for more than 2000 cases and proteomic data for over 700 individuals. Furthermore, we are planning to extend the cohort to cover all patients with hematological malignancies (adding myeloid-lineage malignancies) and adjoining a control population to improve further the usefulness of DALY-CARE.

A recent amendment to the Danish health law enables rapid implementation of research findings with substantial potential to improve patient outcomes, including decision-support tools developed through research platforms such as DALY-CARE.[40] The necessary infrastructure for integrating such models into Danish EHR systems is already established, and has been successfully used to deploy CLL-TIM, a research-based decision-support tool now implemented in clinical practice.[11,13]

Furthermore, ongoing platform enhancements include broader harmonization of EHR data across regions, improved integration of structured and unstructured clinical content, and the deployment of advanced analytics with the aim of optimizing and individualizing patient care.

Data Sharing Statement

Data from the DALY-CARE Genetic Cohort are pseudonymized and subject to data protection regulations. Access may be granted to academic and clinical partners under a Data Processor Agreement (template available https://github.com/RH-CLL-LAB/dalycare_package/tree/main/documentation) and within the scope of the approved DALY-CARE research protocol. Access is limited to collaborative projects within the approved research protocol and conducted in the secure environment at the NGC. Detailed information on access procedures and agreements is provided in the DALY-CARE data resource paper.[10] For inquiries, please contact the corresponding author.

Patient Consent for Publication

Under Danish legislation, register-based studies do not require written informed consent. The National Ethics Committee approved use of retrospective biobank material, including genotyping and other molecular/functional analyses, while written informed consent was obtained for prospective sampling. Patients who wished for their data not to be used for research were removed from this study and the DALY-CARE protocol.

Ethics Approval

The DALY-CARE Genetic cohort has been approved under the DALY-CARE protocol by the Data Protection Agency for the Capital Region and the National Ethics Committee (approvals P-2020-561 and 1804410). In Denmark, collection of nationwide health register data is mandated by law and may be used for research under approved protocols. The National Ethics Committee granted a waiver of informed consent for use of electronic health data and retrospective biobank material, while written informed consent was obtained for prospective biobank sampling. Full details on approvals and consent procedures are provided in the DALY-CARE data resource paper.[10]

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As part of the DALY-CARE data resource, the DALY-CARE genetic cohort is available for collaborative research. Access is granted to academic and clinical partners under a Data Processor Agreement and within the scope of the approved DALY-CARE research protocol. All analyses are performed within the secure computing facilities of the Danish National Genome Center (NGC). We welcome national and international research partners interested in LCs, biomarker discovery, and translational applications in precision medicine, as detailed in the data sharing statement.

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